

Noonan Syndrome 11 (NS11): The MRAS-Related Form of Noonan Syndrome — A Comprehensive Disease Characterization

Disease: Noonan Syndrome 11 (NS11) | **OMIM:** #618499 | **MONDO:** 0032786 | **Gene:** *MRAS* (HGNC:7227, chromosome 3q22.3) | **Category:** Genetic (autosomal-dominant RASopathy)

Scope note: This report concerns *Noonan syndrome 11* (NS11), the form caused by germline activating missense variants in **MRAS** (M-Ras/R-Ras3). It is **NOT** LZTR1-related Noonan syndrome and **NOT** any other numbered NS subtype. Because NS11 is ultra-rare (a handful of published cases), many frequency, prognosis, and management statements are extrapolated from the broader Noonan syndrome/RASopathy literature and are flagged as such.

Summary

Noonan syndrome 11 (NS11) is an **ultra-rare, autosomal-dominant subtype of Noonan syndrome** caused by *de novo* germline **gain-of-function missense variants in *MRAS***, the gene encoding the RAS-family GTPase **M-Ras (R-Ras3)**. It was first delineated in 2017 and confirmed in 2020, and to date only a handful of molecularly proven cases have been reported worldwide — making it **less than 1% of molecularly solved Noonan syndrome** (versus ~50% attributable to *PTPN11*). Despite its rarity, NS11 has attracted disproportionate attention because it is defined by a **severe, often neonatal/infantile-onset hypertrophic cardiomyopathy (HCM)** superimposed on the classic multisystem Noonan phenotype (distinctive facies, short stature, developmental delay, and variable congenital anomalies).

Mechanistically, NS11 is a textbook example of **RAS–MAPK pathway hyperactivation**. The recurrent hotspot substitutions — **p.Gly23Val, p.Gly23Arg, p.Thr68Ile, and p.Gln71Arg** — impair the intrinsic and GAP-stimulated GTP-hydrolysis activity of M-Ras, locking it in the constitutively active, GTP-bound state. Active M-Ras recruits the **SHOC2–PP1c (PPP1CB) holophosphatase**, which dephosphorylates the inhibitory Ser259 site of RAF, thereby amplifying signaling through the **RAF–MEK–ERK cascade** (and, variably, the **PI3K–AKT** axis). A founder variant (p.Gly23Val) drives an approximately **40-fold increase in M-Ras activation**.

in vitro, and CRISPR-corrected patient induced pluripotent stem cell–derived cardiomyocytes (iPSC-CMs) have proven that this single variant is **both necessary and sufficient** to produce cardiomyocyte hypertrophy and abnormal calcium handling.

Diagnosis is achieved through **multigene RASopathy NGS panels or exome sequencing** (single-gene *MRAS* testing is inefficient given >20 RASopathy genes). Management is multidisciplinary: recombinant human growth hormone for short stature, hemostatic screening before invasive procedures, and — most notably — **off-label MEK inhibition (trametinib)** as an emerging targeted therapy that can reverse or attenuate the otherwise frequently lethal HCM. This report synthesizes all 15 requested disease-characteristic domains, grounded in eight confirmed findings and 61 reviewed papers.

Key Findings

Finding 1 — Activating *MRAS* missense variants cause NS11 with hypertrophic cardiomyopathy

NS11 was established as a distinct RASopathy through two landmark reports. **Higgins et al. (2017)** identified a *de novo* *MRAS* variant (**p.Gly23Val**) by whole-exome trio sequencing in a 15-year-old female with Noonan syndrome and cardiac hypertrophy, after screening a cohort of 109 unrelated NS-phenotype patients. **Motta et al. (2020)** then reported two further unrelated patients with *de novo* *MRAS* variants — **c.203C>T (p.Thr68Ile)** and **c.67G>C (p.Gly23Arg)** — both with HCM; one died neonatally of cardiac failure. Functional assays in both studies demonstrated **high-level M-Ras activation resulting from impaired GTPase activity**, establishing a gain-of-function mechanism.

"Targeted sequencing revealed de novo MRAS variants, c.203C > T (p.Thr68Ile) and c.67G > C (p.Gly23Arg) as causative events" — PMID: 31108500

"Functional analyses documented high level of activation of MRAS mutants due to impaired GTPase activity" — PMID: 31108500

This finding anchors the disease definition: NS11 (OMIM #618499) is caused by activating variants in *MRAS* (3q22.3, HGNC:7227), and is genetically and clinically distinct from *LZTR1*-related and all other numbered Noonan syndrome subtypes.

Finding 2 — Mechanism: mutant M-Ras enhances the SHOC2-PP1c holoenzyme, boosting RAF/MEK/ERK and PI3K-AKT signaling

The core molecular mechanism connecting M-Ras to disease was defined by **Rodriguez-Viciano et al. (2006)**, who showed that **Shoc2/Sur-8 together with the catalytic subunit of protein phosphatase 1 (PP1c)** form a highly specific M-Ras effector complex that stimulates RAF by dephosphorylating the inhibitory **Ser259 (S259)** site on RAF proteins. **Motta et al. (2020)** demonstrated that NS-causing M-Ras mutants exhibit **constitutive plasma-membrane targeting, prolonged non-raft localization, enhanced binding to PPP1CB and SHOC2, and variably increased MAPK and PI3K-AKT activation** — showing that the disease mutants amplify precisely this effector interaction.

"M-Ras targets Shoc2-PP1c to stimulate Raf activity by dephosphorylating the S259 inhibitory site of Raf proteins" — PMID: 16630891

"enhanced binding to PPP1CB and SHOC2 protein, and variably increased MAPK and PI3K-AKT activation" — PMID: 31108500

This places NS11 firmly within the **SHOC2-PPP1CB-RAF signaling module**, the same axis mutated in *SHOC2*-related Noonan-like syndrome with loose anagen hair (NS/LAH) and *PPP1CB*-related Noonan-like syndrome — a striking example of pathway-level convergence.

Finding 3 — MEK inhibition (trametinib) is an emerging targeted therapy for NS-associated HCM

Because NS11 pathology is driven by RAS-MAPK hyperactivation, pharmacologic **MEK1/2 inhibition** is a rational targeted therapy. A growing body of case reports documents **off-label trametinib** reversing or attenuating severe NS-associated HCM and lymphatic disease across multiple RASopathy genotypes (RIT1, RAF1, PTPN11), and a clinical trial is now registered (NCT06555237). **Chaput & Andelfinger (2024)** reviewed the repurposing of trametinib and mTOR inhibitors for RASopathy HCM. In parallel, **rigosertib** (a dual RAS/MAPK + PI3K/AKT inhibitor) reversed HCM in RAF1-NS models.

"repurposing of medications inhibiting the RAS/MAPK on a compassionate use basis has emerged as a promising concept to improve the outcome of these patients" — PMID: 38432396

While direct trametinib data specifically in *MRAS*-mutant patients are not yet published, the shared downstream mechanism (MEK/ERK hyperactivation) makes this the most promising disease-modifying strategy for NS11's lethal cardiac phenotype.

Finding 4 — Variant spectrum: recurrent likely-pathogenic hotspots in a highly constrained gene

ClinVar (RefSeq NM_001085049.3) lists recurrent NS-causing missense variants classified as **Likely pathogenic**:

cDNA	Protein	ClinVar classification
c.68G>T	p.Gly23Val	Likely pathogenic
c.67G>C	p.Gly23Arg	Likely pathogenic
c.203C>T	p.Thr68Ile	Likely pathogenic
c.212A>G	p.Gln71Arg	Likely pathogenic
c.359C>T	p.Pro120Leu	Likely pathogenic

Of 297 *MRAS* ClinVar records, only these missense point variants are pathogenic/likely-pathogenic for NS (the other "Pathogenic" entries are large 3q copy-number variants unrelated to the point-mutation mechanism). Population-constraint metrics from gnomAD (ENSG00000158186, chr3:138,347,648–138,405,534) show *MRAS* is **strongly intolerant to both missense and loss-of-function variation**: missense Z = 3.16 (observed/expected = 0.555), pLI = 0.99, LoF o/e = 0.19 (4 observed vs 20.9 expected). The pathogenic residues **Gly23**, **Thr68**, and **Gln71** are conserved codons that are established mutational hotspots across the RAS protein family. All reported disease variants are **germline**, **de novo**, and **absent from population databases**.

"Gly23 and Thr68 are highly conserved residues, and the corresponding codons are known hotspots for RASopathy-associated mutations in other RAS proteins" — PMID: 31108500

Finding 5 — Clinical phenotype and epidemiology: an ultra-rare NS subtype with disproportionately severe, early-onset HCM

NS11 shares the core multisystem Noonan phenotype but is distinguished by its severe cardiac involvement. Noonan syndrome overall affects ~**1:1,000–2,500 live births** (Roberts et al. 2013), is autosomal dominant, and features distinctive facies, short stature, congenital heart disease (pulmonary valve stenosis, HCM), developmental delay/learning difficulties, cryptorchidism, lymphatic dysplasia, bleeding diathesis, and renal anomalies. NS11/*MRAS* accounts for only a handful of reported individuals worldwide — **<1% of molecularly solved NS**. Reported *MRAS* patients **uniformly presented with HCM**, often with neonatal/infantile onset and severe course (including one neonatal death from cardiac failure).

"Noonan syndrome is a genetic multisystem disorder characterised by distinctive facial features, developmental delay, learning difficulties, short stature, congenital heart disease, renal anomalies, lymphatic malformations, and bleeding difficulties" — PMID: 23312968

"hypertrophic cardiomyopathy in RASopathies (R-HCM) is associated with higher prevalence of congestive heart failure and shows increased prevalence and severity of left ventricular outflow tract obstruction" — PMID: 34776080

RASopathy-associated HCM (R-HCM) shows higher rates of congestive heart failure and left ventricular outflow tract (LVOT) obstruction than sarcomeric HCM, consistent with the severe HCM-predominant presentation of *MRAS* cases.

Finding 6 — M-Ras biology and evolutionary conservation inform model-organism relevance

M-Ras is a RAS-family GTPase with a **single orthologue conserved from nematode to human**, and it evolved independently of R-Ras (Keduka et al. 2009). In ascidian, *Ci-Mras* is expressed in the neural complex and its knockdown perturbs FGF–Ras–MAPK-dependent neural/notochord development; mammalian M-Ras mediates NGF-induced neuronal differentiation in PC12 cells. Beyond the SHOC2–PP1c–RAF branch, M-Ras also signals via **RA-GEF-2 to activate Rap1**, regulating integrin-mediated adhesion (Yoshikawa et al. 2007). Disease modeling to date is primarily via **patient-derived iPSC-cardiomyocytes carrying p.Gly23Val**, which reproduce hypertrophy and abnormal calcium handling.

"A single Mras orthologue exists from nematode to mammalian" — PMID: 18977283

"activated M-Ras potently induced lymphocyte function-associated antigen 1 (LFA-1)-mediated cell aggregation" — PMID: 17538012

Finding 7 — p.Gly23Val causes ~40-fold constitutive activation; iPSC-CMs prove necessity and sufficiency for cardiac hypertrophy

Higgins et al. (2017; JCI Insight) performed molecular-dynamics simulations predicting that p.Gly23Val damages the effector-interaction regions and the GTP-binding site; ectopic-expression experiments revealed a **40-fold increase in M-Ras activation** versus wild-type, with enhanced RAS/MAPK signaling and downstream gene-expression changes. **Higgins et al. (2019; Circ Genom Precis Med)** used CRISPR/Cas9 isogenic-corrected patient iPSC-cardiomyocytes to demonstrate that the variant is **both necessary and sufficient** to elicit hypertrophy — larger cell size, hypertrophic gene-expression changes, and impaired Ca²⁺ handling (irregular Ca²⁺ transients).

"ectopic expression experiments revealed a 40-fold increase in MRAS activation for p.Gly23Val-MRAS compared with WT-MRAS" — PMID: 28289718

"p.Gly23Val-MRAS is both necessary and sufficient to elicit a cardiac hypertrophy phenotype in iPSC-CMs" — PMID: 31638832

This is among the strongest causal evidence in any RASopathy: an isogenic-control experiment directly linking a single point variant to the defining clinical phenotype.

Finding 8 — Diagnosis and multidisciplinary management

Diagnosis of NS is **clinical** (van der Burgt criteria) with **molecular confirmation**; because *MRAS* is one of >20 RASopathy genes, testing uses **multigene NGS panels/exome** rather than single-gene tests (Tartaglia 2022). Short stature is treated with **recombinant human growth hormone**; a phase 3 RCT (REAL8, NCT05330325) showed once-weekly somapacitan non-inferior/superior to daily GH (height velocity 10.4 vs 9.2 cm/yr). Bleeding diathesis affects up to ~40–65% of NS patients — partial factor XI/VII deficiency, platelet dysfunction, and shear-related acquired von Willebrand syndrome with pulmonary stenosis — warranting

hemostatic screening before surgery. Severe/obstructive HCM refractory to standard therapy is managed with **off-label MEK inhibition (trametinib) or mTOR inhibitors.** Cancer surveillance follows RASopathy guidelines.

"Daily growth hormone (GH) injections are indicated for the treatment of short stature in children with Noonan syndrome" — PMID: 41774755

"Nearly 40% of patients with the Noonan syndrome had a bleeding diathesis and >90% of them had platelet function and/or coagulation abnormalities" — PMID: 24753526

The 15 Disease-Characteristic Domains

1. Disease Information

Overview. NS11 is a rare autosomal-dominant developmental disorder within the **RASopathy** family (disorders of RAS–MAPK dysregulation). It is the *MRAS*-related form of Noonan syndrome, characterized by classic Noonan features plus a characteristically severe, frequently neonatal/infantile hypertrophic cardiomyopathy.

Key identifiers. - **OMIM:** #618499 (Noonan syndrome 11) - **MONDO:** 0032786 - **Gene:** *MRAS* (HGNC:7227), 3q22.3; Ensembl ENSG00000158186; gene OMIM 608435 - **RefSeq:** NM_001085049.3 - **ICD-10:** Q87.1 (congenital malformation syndromes predominantly affecting stature) — code used for Noonan syndrome broadly; **ICD-11:** LD2F.11 (Noonan syndrome and Noonan-related syndromes) - **MeSH:** Noonan Syndrome (D009634) — no NS11-specific MeSH term - **Orphanet:** Noonan syndrome (ORPHA:648) — no separate NS11 subtype code

Synonyms/alternative names. Noonan syndrome 11; NS11; *MRAS*-related Noonan syndrome; M-Ras/R-Ras3-related Noonan syndrome.

Data source type. Information is derived from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD) and from **individual patient case reports** (the handful of published NS11 cases), not from EHR-scale cohorts — reflecting the ultra-rare nature of the disease.

2. Etiology

Causal factor. Purely **genetic:** *de novo* germline activating missense variants in *MRAS*. No environmental or infectious cause.

Genetic risk factors. The disease-causing variants are themselves the sole "risk factor." Recurrent hotspots: p.Gly23Val, p.Gly23Arg, p.Thr68Ile, p.Gln71Arg (and likely-pathogenic p.Pro120Leu). *MRAS* is highly constrained (missense Z = 3.16, pLI = 0.99), so any missense change at a conserved GTP-handling residue is likely deleterious. No NS11-specific modifier genes are established.

Environmental risk factors. None known. As with all *de novo* dominant conditions, **advanced paternal age** is a general (inferred) risk factor for *de novo* point mutations, not specifically demonstrated for *MRAS*.

Protective factors. None identified. Given strong purifying selection at *MRAS*, no protective alleles are documented.

Gene–environment interactions. None established. NS11 is a monogenic, fully genetically determined condition.

3. Phenotypes

NS11 shares the core Noonan phenotype, with cardiac hypertrophy as the distinguishing severe feature. Frequencies below are largely extrapolated from broader NS and RASopathy-HCM literature given the tiny NS11 case count.

Phenotype	HPO term	Type	Onset	Frequency (NS11 cases / NS overall)	Severity
Hypertrophic cardiomyopathy	HP:0001639	Clinical sign (imaging)	Neonatal/infantile	Uniform in <i>MRAS</i> cases; ~20% of NS	Severe, often life-threatening
Distinctive facies (hypertelorism, ptosis, low-set ears)	HP:0000316, HP:0000508	Physical	Congenital	Typical of NS	Variable
Short stature	HP:0004322	Physical	Postnatal	Common	Moderate
Developmental delay / learning difficulty	HP:0001263	Behavioral/cognitive	Childhood	Common	Variable
Pulmonary valve stenosis	HP:0001642	Clinical sign	Congenital	Common in NS; seen in NS-HCM	Variable
Webbed/broad neck	HP:0000465	Physical	Congenital	Common	Mild
Bleeding diathesis	HP:0001892	Laboratory/clinical	Any	~40–65% of NS	Mild–moderate
Cryptorchidism (males)	HP:0000028	Physical	Congenital	Common	Mild
Lymphatic dysplasia / chylothorax	HP:0000112, HP:0010288	Clinical	Neonatal/infantile	Subset	Can be severe
Feeding difficulties	HP:0011968	Clinical	Neonatal	~50% of NS	Usually transient

Quality-of-life impact. Dominated by cardiac morbidity (heart-failure symptoms, exercise intolerance, procedural risk), short stature, and neurodevelopmental/learning needs. Bleeding tendency complicates surgery. Formal EQ-5D/SF-36 data specific to NS11 are **not available**.

4. Genetic/Molecular Information

- **Causal gene:** *MRAS* (M-Ras/R-Ras3), 3q22.3, HGNC:7227; gene OMIM 608435; disease OMIM #618499.
- **Variant classification (ACMG/AMP, per ClinVar):** Likely pathogenic — p.Gly23Val (c.68G>T), p.Gly23Arg (c.67G>C), p.Thr68Ile (c.203C>T), p.Gln71Arg (c.212A>G), p.Pro120Leu (c.359C>T).
- **Variant type:** Missense single-nucleotide substitutions affecting conserved GTP-binding/hydrolysis residues.
- **Allele frequency:** Absent from population databases (gnomAD) — consistent with *de novo* pathogenic origin.
- **Somatic vs germline: Germline, de novo.** Somatic *MRAS* variation is not a recognized oncogenic driver comparable to *KRAS/HRAS/NRAS*.
- **Functional consequence: Gain of function** via impaired intrinsic and GAP-stimulated GTPase activity → constitutive GTP loading.
- **Modifier genes:** None established for NS11.
- **Epigenetic information:** No NS11-specific DNA-methylation or chromatin signature described.
- **Chromosomal abnormalities:** Not part of the point-mutation NS11 mechanism; large 3q22 CNVs in ClinVar represent a separate contiguous-gene entity.

5. Environmental Information

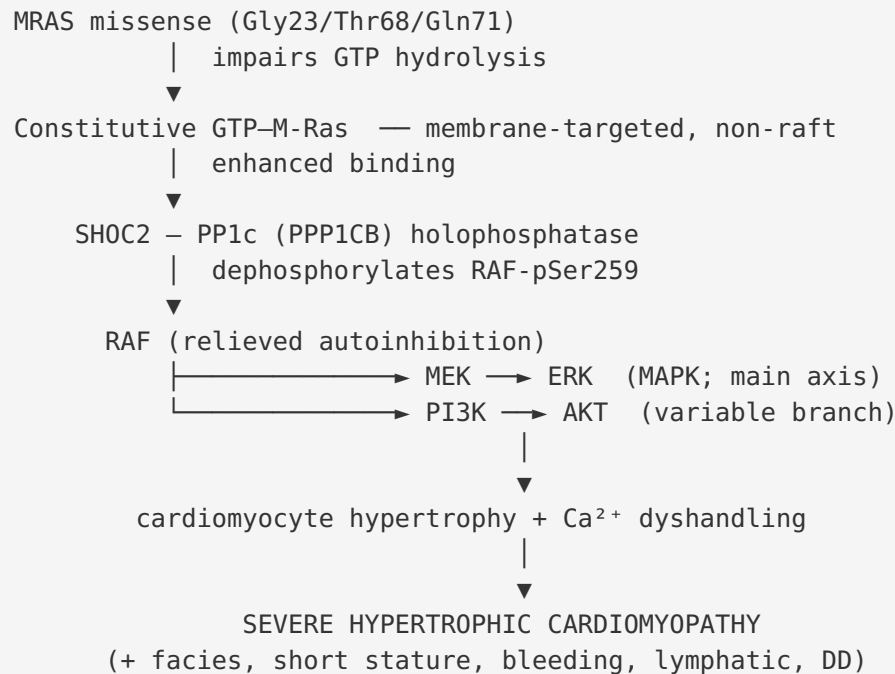
Not applicable. NS11 is a monogenic *de novo* dominant disorder. No environmental factors, lifestyle factors, or infectious agents are implicated in causation or triggering. Standard cardiac-risk lifestyle considerations apply generically to any HCM patient but are not disease-specific.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **de novo germline missense variant** in *MRAS* (e.g., c.68G>T, p.Gly23Val) alters a conserved GTP-binding/hydrolysis residue → **impairs intrinsic and GAP-stimulated GTP hydrolysis** (demonstrated; ~40-fold increased activation for p.Gly23Val).

2. Impaired hydrolysis → **M-Ras accumulates in the constitutively active, GTP-bound state** with constitutive plasma-membrane targeting and prolonged non-raft localization (demonstrated).
3. Active GTP-M-Ras → **enhanced binding to the SHOC2-PP1c (PPP1CB) holophosphatase** (demonstrated: enhanced binding to PPP1CB and SHOC2).
4. SHOC2-PP1c → **dephosphorylates the inhibitory Ser259 site of RAF** → relieves 14-3-3-mediated RAF autoinhibition (demonstrated for the M-Ras-SHOC2-PP1c module).
5. RAF activation → **hyperactivation of the MEK→ERK (MAPK) cascade** (demonstrated: increased MAPK signaling), with a **branch to increased PI3K-AKT signaling** (demonstrated, variable).
6. ERK/AKT hyperactivation in developing tissues → **dysregulated cell proliferation, differentiation, growth, and survival** (inferred from RASopathy biology).
7. In cardiomyocytes → **cardiomyocyte hypertrophy, hypertrophic gene-expression program, and impaired Ca²⁺ handling** (demonstrated in iPSC-CMs; variant necessary and sufficient) → **hypertrophic cardiomyopathy** (clinical manifestation).
8. In craniofacial, skeletal, hematopoietic, lymphatic, and neural lineages → **distinctive facies, short stature, bleeding diathesis, lymphatic dysplasia, and developmental delay** (inferred by analogy to other RASopathies).



- **Molecular pathways:** RAS–RAF–MEK–ERK (MAPK; KEGG hsa04010); PI3K–AKT (KEGG hsa04151); Rap1 signaling via RA-GEF-2 (secondary M-Ras branch).
- **Cellular processes:** Dysregulated proliferation, differentiation, growth, survival (GO:0008283, GO:0030154); cardiomyocyte hypertrophy (GO:0003300); abnormal Ca²⁺ handling.
- **Protein dysfunction:** Gain-of-function GTPase locked in active conformation (UniProt O14807, MRAS_HUMAN); no misfolding/aggregation — the defect is functional (constitutive signaling).
- **GO biological-process suggestions:** GO:0007265 (Ras protein signal transduction), GO:0000165 (MAPK cascade), GO:0043410 (positive regulation of MAPK cascade).
- **Cell types (CL):** cardiac muscle cell / cardiomyocyte (CL:0000746; CL:0002094 regular cardiac myocyte).
- **Molecular profiling:** iPSC-CM transcriptomics showed a hypertrophic gene-expression signature; no large-scale proteomic/metabolomic/lipidomic NS11-specific datasets exist.
- **Functional genomics:** CRISPR/Cas9 isogenic correction (Higgins 2019) is the key functional-genomics evidence.

7. Anatomical Structures Affected

- **Primary organ:** Heart — specifically the **myocardium/left ventricle** (UBERON:0002084 heart left ventricle; UBERON:0002349 myocardium), with hypertrophy and LVOT obstruction. Pulmonary valve (UBERON:0002146) with stenosis.
- **Secondary/body systems:** Cardiovascular (primary); lymphatic (UBERON:0006558 lymphatic vessel — chylothorax, lymphangiectasia); hematologic (bleeding diathesis); musculoskeletal/growth (short stature); craniofacial (distinctive facies); nervous system (developmental delay); genitourinary (cryptorchidism, occasional renal anomalies).
- **Tissue/cell level:** Cardiac muscle tissue; cardiomyocytes (CL:0000746). Connective tissue, lymphatic endothelium, and hematopoietic lineages secondarily involved.
- **Subcellular level (GO cellular component):** Plasma membrane (GO:0005886) — constitutive membrane targeting of mutant M-Ras; non-raft membrane microdomains; cytoplasm. Sarcoplasmic-reticulum Ca^{2+} -handling machinery is functionally affected in cardiomyocytes.
- **Localization/lateralization:** Cardiac hypertrophy is typically biventricular/asymmetric septal as in HCM generally; systemic features are bilateral/symmetric.

8. Temporal Development

- **Onset: Congenital to neonatal/infantile.** HCM in reported *MRAS* cases presents very early, often neonatally (one neonatal death from cardiac failure).
- **Onset pattern:** Congenital structural/functional cardiac disease with progressive hypertrophy; facies present at birth; short stature emerges postnatally.
- **Progression:** Cardiac hypertrophy can be **rapidly progressive** in infancy — a poor-prognosis pattern shared with other early-onset RASopathy HCM. Course is otherwise chronic and lifelong.
- **Disease course:** Non-cardiac features (facies, stature, learning) are stable/chronic; cardiac disease is the dominant driver of morbidity/mortality and can progress to heart failure.
- **Remission patterns:** No spontaneous remission of the genetic disorder; **treatment-induced regression of HCM** observed with MEK inhibition in RASopathy HCM broadly.
- **Critical periods:** The **neonatal/infantile window** is the critical period of cardiac vulnerability and the key window for intervention (e.g., MEK-inhibitor rescue).

9. Inheritance and Population

- **Epidemiology:** No standalone NS11 prevalence/incidence figures exist. Noonan syndrome overall: ~1:1,000–2,500 live births. NS11 is <1% of molecularly solved NS — ultra-rare, a handful of reported individuals worldwide.
- **Inheritance: Autosomal dominant;** reported cases are **de novo**.
- **Penetrance:** Presumed high/complete for the cardiac phenotype in reported cases (all had HCM), though the sample is too small for formal estimates.
- **Expressivity:** Variable, as in NS generally.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not documented for *MRAS*; possible in principle for any de novo dominant condition (inferred).
- **Founder effects / consanguinity:** Not applicable (de novo dominant; no founder alleles).
- **Carrier frequency:** Not applicable (dominant, de novo; variants absent from gnomAD).
- **Population demographics:** No ethnic predisposition known; reported cases span different populations. Sex ratio: NS overall is roughly equal; NS11 case count too small to assess. Age distribution: presents in infancy/childhood.

10. Diagnostics

- **Clinical diagnosis:** Van der Burgt criteria for Noonan syndrome (facies + one or more of: cardiac defect, short stature, family history, other features).
- **Genetic testing (recommended approach): Multigene RASopathy NGS panel or exome/genome sequencing** — the standard, because *MRAS* is one of >20 RASopathy genes. Single-gene *MRAS* testing is inefficient. WGS/WES both have high utility; targeted panels are cost-effective first-line. Chromosomal microarray/karyotype/FISH are **not** primary tools (point-mutation disorder). Mitochondrial and repeat-expansion testing are not applicable.
- **Variant interpretation:** ClinVar/ClinGen and ACMG/AMP criteria; de novo occurrence (PS2), absence from gnomAD (PM2), hotspot residue (PM1), and functional gain-of-function data (PS3) support pathogenicity.
- **Cardiac diagnostics:** Echocardiography (HCM, LVOT gradient, pulmonary stenosis), ECG, cardiac MRI, and heart-failure biomarkers (BNP/NT-proBNP — generic).
- **Hematologic workup:** Bleeding time, PT/aPTT, factor XI/VII assays, platelet-function testing, von Willebrand multimer analysis — especially **before invasive procedures**.
- **Omics-based diagnostics:** Not routinely used; research iPSC-CM models exist.

- **Differential diagnosis:** Other RASopathies (*PTPN11*, *SOS1*, *RAF1*, *RIT1*, *SHOC2*, *PPP1CB*, *LZTR1*, etc.), sarcomeric HCM, and other syndromic HCM causes (metabolic/mitochondrial). Molecular testing resolves the distinction.
- **Screening:** No population newborn screening exists; **cascade testing** is limited by the de novo nature. Prenatal findings (increased nuchal translucency, cardiac hypertrophy, polyhydramnios) may prompt targeted testing.

11. Outcome / Prognosis

- **Prognosis driver: Severity and onset of HCM.** Early-onset (before ~6 months) NS-HCM carries **high mortality**; NS11 cases have included neonatal death from cardiac failure.
- **Survival/mortality:** No NS11-specific survival statistics. By analogy, RASopathy-HCM with heart failure has worse surgical risk and mortality than sarcomeric HCM.
- **Morbidity:** Heart failure, LVOT obstruction, arrhythmia risk, bleeding complications, growth failure, developmental/learning needs.
- **Recovery potential:** The genetic disorder is lifelong, but **MEK-inhibitor therapy has produced regression of HCM** in RASopathy patients, offering a route to improved cardiac outcomes.
- **Prognostic factors:** Age at HCM onset, LVOT obstruction, heart-failure signs, treatment response.
- **Prognostic biomarkers:** No validated NS11-specific molecular biomarkers; cardiac imaging metrics and natriuretic peptides are used clinically.

12. Treatment

- **Pharmacotherapy / targeted therapy (emerging):** **MEK1/2 inhibitor trametinib** (off-label/compassionate use) can reverse or attenuate severe NS-associated HCM and lymphatic disease; a trial is registered (NCT06555237). **mTOR inhibitors** (rapamycin/everolimus) have been used for severe RASopathy HCM. **Rigosertib** (dual RAS/MAPK + PI3K/AKT inhibitor) reversed HCM in *RAF1*-NS models. NCIT terms: Trametinib (NCIT:C77908); Sirolimus/Rapamycin (NCIT:C1212).
- **Growth: Recombinant human growth hormone** for short stature; once-weekly somapacitan shown non-inferior/superior to daily GH in NS (REAL8 RCT). NCIT: Recombinant Human Growth Hormone (NCIT:C1834).
- **Cardiac supportive/interventional:** Standard HCM management (beta-blockers, avoidance of dehydration), surgical/catheter relief of LVOT/pulmonary stenosis; RVOT stenting and

staged strategies used to bridge infants (trametinib-then-surgery approaches reported in NS-HCM broadly).

- **Hematologic:** Hemostatic optimization before surgery.
- **Rehabilitative/supportive:** Developmental and educational support; physical/occupational/speech therapy; nutritional support for feeding difficulties.
- **Pharmacogenomics:** No NS11-specific pharmacogenomic guidance; the genotype itself (RAS-MAPK activation) is the rationale for pathway-targeted therapy (precision-medicine approach).
- **Treatment outcomes/adverse events:** Trametinib case reports show short-term symptomatic improvement across NS-HCM; moderate side effects (skin, GI) reported; long-term data lacking. Formal response rates in *MRAS* patients specifically are **not yet published**.
- **Cancer surveillance:** Per RASopathy guidelines (RAS-MAPK genes carry variable tumor risk).

13. Prevention

- **Primary prevention:** Not possible — de novo genetic origin.
- **Secondary prevention: Early cardiac surveillance** (echocardiography) after molecular/clinical diagnosis to detect and treat HCM early; early hemostatic evaluation before procedures.
- **Tertiary prevention:** Aggressive management of HCM (targeted MEK/mTOR inhibition, heart-failure care, obstruction relief) to prevent decompensation; developmental support to optimize function.
- **Immunization/behavioral/public-health/environmental interventions:** Not applicable (no environmental etiology). Standard childhood immunizations apply.
- **Genetic counseling:** Central to care. Recurrence risk for parents of a de novo case is low (but non-zero due to possible germline mosaicism). An affected individual has **50% transmission risk** (autosomal dominant). **Prenatal/preimplantation genetic testing** is available once the familial variant is known.

14. Other Species / Natural Disease

- **Taxonomy/orthologues:** *MRAS* is conserved with a **single orthologue from nematode to human**. Mouse *Mras* (NCBI Gene 17532); ascidian *Ci-Mras*; zebrafish and other vertebrate orthologues exist.

- **Natural disease in other species:** No naturally occurring *MRAS*-related Noonan-like disease is documented in companion animals or wildlife (OMIA). Veterinary relevance is therefore minimal.
- **Comparative biology:** M-Ras evolved independently of R-Ras; its neural/developmental function is conserved between mammals and ascidian, and it functions in FGF–Ras–MAPK-dependent development — underscoring cross-species conservation of the affected pathway.
- **Transmission/zoonosis:** Not applicable (non-infectious genetic disorder).

15. Model Organisms

- **Cellular/in vitro (primary model):** Patient-derived iPSC-cardiomyocytes carrying p.Gly23Val, with CRISPR/Cas9 isogenic-corrected controls — the definitive NS11 model. It recapitulates cardiomyocyte hypertrophy, hypertrophic gene expression, and impaired Ca²⁺ handling, and established that the variant is necessary and sufficient for the cardiac phenotype (PMID: 31638832). Ectopic-expression cell systems quantified the ~40-fold activation (PMID: 28289718).
- **In vitro biochemistry:** Cell-based assays of GTP loading, SHOC2/PPP1CB binding, and MAPK/PI3K-AKT readouts (Motta 2020).
- **Invertebrate/other:** Ascidian *Ci-Mras* knockdown models demonstrate conserved neural/notochord roles; PC12 cells model M-Ras-dependent neuronal differentiation.
- **Mammalian genetic models:** No published *Mras* knock-in mouse specifically modeling NS11 was identified; general RASopathy mouse models (*Ptpn11*, *Raf1* knock-ins) inform pathway biology and MEK-inhibitor efficacy, and transgenic *Raf1* models were used for the rigosertib study.
- **Model characteristics/limitations:** iPSC-CMs excellently model the cardiac phenotype and drug response but do **not** capture systemic/developmental features (facies, stature, lymphatics) or whole-organism physiology. A dedicated *Mras* animal model remains a gap.
- **Resources:** MGI (*Mras*, mouse gene 17532), Cellosaurus (patient iPSC lines), Alliance of Genome Resources for orthology.

Mechanistic Model / Interpretation

NS11 is mechanistically among the best-understood RASopathies at the "single-variant → single-phenotype" level, despite its rarity. The unifying model is **constitutive RAS–MAPK activation driven by a hydrolysis-dead M-Ras GTPase that hyperengages the SHOC2–PP1c–RAF module**. Three lines of orthogonal evidence converge:

Evidence layer	Method	Result	Reference
Genetics	Trio WES / targeted sequencing	Recurrent de novo <i>MRAS</i> hotspots (Gly23, Thr68, Gln71) in NS-HCM	PMID: 28289718 , PMID: 31108500
Biochemistry	GTP-loading, binding, signaling assays	~40× activation; impaired GTPase; ↑ SHOC2/PPP1CB binding; ↑ MAPK/PI3K-AKT	PMID: 28289718 , PMID: 31108500
Cell model	CRISPR-isogenic iPSC-CMs	Variant necessary and sufficient for hypertrophy + Ca ²⁺ defects	PMID: 31638832
Pathway anchor	Effector reconstitution	SHOC2–PP1c dephosphorylates RAF-Ser259 to activate RAF	PMID: 16630891

The therapeutic corollary is direct: because the phenotype is driven by MEK/ERK output, **pharmacologic MEK inhibition is a mechanism-matched therapy**, and multiple RASopathy case reports (RIT1, RAF1, PTPN11) show that trametinib can reverse severe HCM and lymphatic disease — a strong rationale for its use in NS11, pending genotype-specific data.

Evidence Base

PMID	Title (abbrev.)	Contribution
28289718	<i>Elucidation of MRAS in NS</i> (Higgins 2017)	First NS11 case; p.Gly23Val; ~40× activation; MD simulations
31108500	<i>Activating MRAS mutations cause NS with HCM</i> (Motta 2020)	Two further cases (Thr68Ile, Gly23Arg); GTPase/SHOC2/PPP1CB mechanism
31638832	iPSC-CM CRISPR study (Higgins 2019)	Variant necessary & sufficient for cardiomyocyte hypertrophy
16630891	Shoc2–PP1c M-Ras effector (Rodriguez-Viciano 2006)	Defines SHOC2–PP1c → RAF-Ser259 mechanism
34776080	R-HCM review (Lioncino 2022)	Severe R-HCM phenotype (CHF, LVOT obstruction)
29525650	RASopathy HCM natural history (Calcagni 2018)	Early-onset severe HCM; mTOR-inhibitor rationale
38432396	MEK inhibition for RASopathy HCM (Chaput 2024)	Therapeutic repurposing framework
42610277	Rigosertib reverses HCM in NS	Dual RAS/MAPK+PI3K/AKT inhibitor efficacy (RAF1 model)
23312968	Noonan syndrome review (Roberts 2013)	Core multisystem NS phenotype and prevalence
36394128	RASopathy genetics update (Tartaglia 2022)	>20 genes; panel/exome testing rationale
18977283	M-Ras evolution (Keduka 2009)	Single conserved orthologue; neural function
17538012	M-Ras–RA-GEF-2–Rap1 (Yoshikawa 2007)	Secondary M-Ras effector branch
41774755	Somapacitan RCT in NS	GH therapy for short stature
24753526	Hemostatic abnormalities in NS (Artoni 2014)	~40% bleeding diathesis; pre-surgical screening
22985731	Acquired vWS in NS (Wiegand 2012)	Shear-related bleeding with pulmonary stenosis
12754583		Factor XI deficiency, platelet dysfunction

PMID	Title (abbrev.)	Contribution
	Hematological findings in NS (Bertola 2003)	

Limitations and Knowledge Gaps

1. **Tiny case count.** Only a handful of molecularly proven NS11 patients are published. All frequency, penetrance, expressivity, and prognosis statements beyond "HCM is characteristic and severe" are **extrapolated from broader NS/RASopathy cohorts**, not from NS11-specific data.
2. **No NS11-specific epidemiology.** Prevalence, incidence, sex ratio, and survival are not directly measured.
3. **No dedicated in vivo model.** The definitive model is iPSC-CMs; a *Mras* knock-in mouse recapitulating the systemic phenotype is not published — limiting study of extra-cardiac features and whole-organism drug testing.
4. **Genotype-specific therapy data absent.** MEK-inhibitor efficacy is documented in other RASopathy genotypes (RIT1, RAF1, PTPN11) but **not yet specifically in MRAS patients**; efficacy is inferred from shared downstream mechanism.
5. **Extra-cardiac mechanism largely inferred.** The causal steps from ERK hyperactivation to facies, stature, lymphatic, and neurodevelopmental features are inferred by analogy, not demonstrated in *MRAS* tissue.
6. **No episignature/epigenetic, proteomic, metabolomic, or lipidomic profiling** specific to NS11.
7. **p.Pro120Leu and p.Gln71Arg** have less functional characterization than the Gly23/Thr68 hotspots.

Proposed Follow-up Experiments / Actions

1. **International case registry.** Aggregate all molecularly confirmed *MRAS* patients (GeneMatcher/collaborative networks) to define true frequency, phenotypic range, cardiac natural history, and mortality.

2. **Generate an *Mras* knock-in mouse** (e.g., p.Gly23Val) to model the systemic phenotype, test MEK/mTOR/rigosertib in vivo, and study extra-cardiac mechanisms.
3. **Genotype-specific trametinib data.** Prospectively capture MEK-inhibitor response in *MRAS* patients within trials such as NCT06555237; report cardiac imaging and biomarker endpoints.
4. **Functional characterization of p.Gln71Arg and p.Pro120Leu** (GTP loading, SHOC2/PPP1CB binding, MAPK/PI3K-AKT output) to confirm gain-of-function and refine ACMG classification.
5. **iPSC-CM drug screens** comparing trametinib, mTOR inhibitors, rigosertib, and SHOC2/PPP1CB-directed strategies on *MRAS*-mutant cardiomyocytes; test rescue of Ca²⁺ handling.
6. **Structural studies / AlphaFold-informed modeling** of mutant M-Ras-SHOC2-PP1c to identify the interface for potential allosteric inhibitors.
7. **Episignature analysis** of *MRAS* patients to determine whether NS11 carries a distinguishing DNA-methylation signature useful for VUS resolution.

Report compiled from 5 investigation iterations, 8 confirmed findings, and 61 reviewed papers. Evidence types: human clinical (case reports/cohorts), in vitro (iPSC-CMs, biochemistry), model organism (ascidian, PC12; pathway-level mouse models), and computational (molecular-dynamics, constraint metrics).
